UEC Hybrid Simulator — Decision Report

Model Emergency Department (Type 1)

Executive summary

- Modelled demand: 329 attendances/day (1,973 over the simulated period).
- 4-hour standard: 77% (95% CI 75-80% over 16 weeks); 446 breaches.
- 12-hour waits 1.1%; admitted 15%; mean LOS 2.9 h.
- Inpatient bed occupancy 75% — the System-Dynamics driver of ED exit-block.

Bottleneck diagnosis

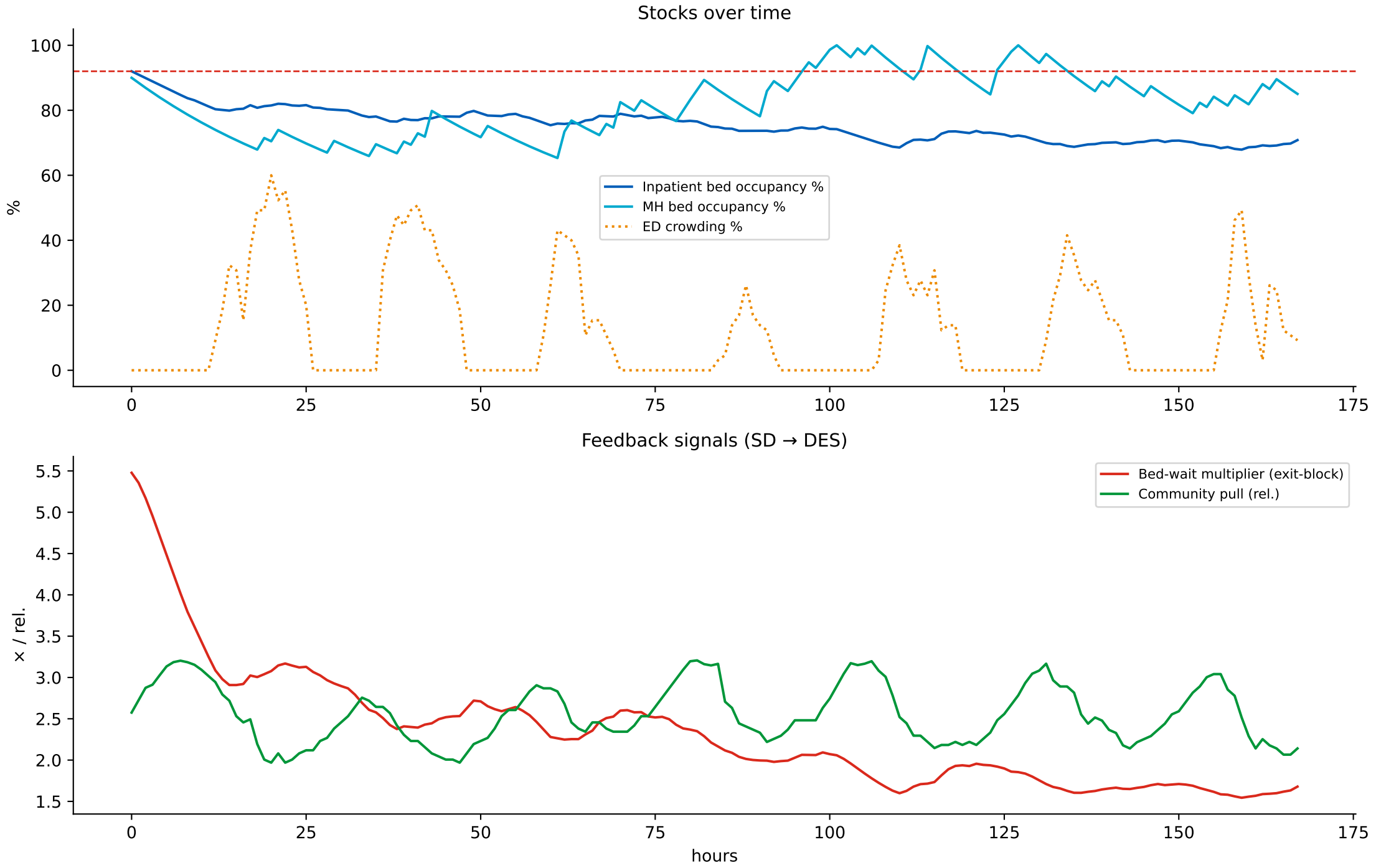
Biggest delay: ED ambulatory (fit to sit) — mean wait 41 min (252 patient-hours lost).

Recommendation: ed doctor is busy (64% utilisation) and likely drives this queue.

UEC scorecard vs national standards

4-hour standard: 77% (target ≥78%)	**AMBER**
Initial assessment ≤15 min: 89% (target ≥90%)	**AMBER**
12-hour waits: 1% (target <10%)	**GREEN**
Ambulance handover ≤45 min: 100% (target ≥95%)	**GREEN**
MH liaison ≤1 hour: 100% (target ≥90%)	**GREEN**
Mental health >24h in ED: 4% (target <5%)	**GREEN**
Children's 4-hour: 86% (target ≥80%)	**GREEN**
EEMAC conversion to admission: 11% (target <20%)	**GREEN**
Inpatient bed occupancy: 75% (target <92%)	**GREEN**

What-if scenarios

Scenario	4-hour	vs baseline	bed occ
Baseline	79%	—	74%
+40 inpatient beds	78%	**-0 pts**	76%
+30% community capacity	81%	**+2 pts**	71%
Winter +15% demand	73%	**-6 pts**	79%

System Dynamics — whole-system stocks and feedback